

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	Gayathri P	Reg. No.:	192424324
Section / Batch:	AIDS-2024	Date:	

Code of Conduct

I Gayathri P(192424324) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Gayathri P

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3

Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4
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Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

Answer:

1. Action–Object Relationships

The semantic representations follow an **action + object + customer** structure:

Customer Query	Action	Object	Relationship
Activate international roaming	ACTIVATE	Roaming	Customer activates Roaming
Deactivate caller tune	DEACTIVATE	CallerTune	Customer deactivates CallerTune
Check data balance	QUERY	DataBalance	Customer queries DataBalance
Enable 5G service	ACTIVATE	5GService	Customer activates 5GService

The general semantic pattern is:

ACTION → OBJECT → CUSTOMER

For example, ACTIVATE(Roaming, Customer) means the Customer wants to activate the Roaming service.

2. Query with Semantic Interpretation Error

The incorrect prediction is Q2.

- Actual Intent: Deactivate Caller Tune
- Predicted Intent: Activate Caller Tune

The semantic representation should be:

DEACTIVATE(CallerTune, Customer)

But the chatbot interpreted the action as ACTIVATE instead of DEACTIVATE.

This is an action-level semantic error, because the object *CallerTune* was identified correctly, but the operation was reversed.

3. Effect on Chatbot Decision Accuracy

Meaning representation directly affects chatbot accuracy.

- Correct semantic representation → correct intent → correct action.
- Incorrect semantic representation → wrong intent → wrong customer service action.

Here, Q1, Q3, and Q4 are correctly interpreted, while Q2 is incorrect.

$$\text{Accuracy} = 3/4 \times 100 = 75\%$$

Therefore, the chatbot achieved 75% intent accuracy for these four queries.

The Q2 error could cause an actual service problem because the system might activate a caller tune when the customer actually requested deactivation.

4. Recommended Improvements

1. **Improve action detection** – Clearly distinguish words such as *activate*, *enable*, *deactivate*, *disable*, and *cancel*.
2. **Use context analysis** – Consider the complete sentence instead of relying only on individual keywords.
3. **Handle negation carefully** – Words such as *not*, *don't*, *deactivate*, and *disable* can completely change the meaning.

4. **Use domain-specific training data** – Train the semantic model with more telecom-related queries.
5. **Add intent validation** – Verify that the detected action and object form a valid telecom operation before executing it.
6. **Use confidence thresholds** – If the model is uncertain, ask the customer for confirmation instead of performing a potentially incorrect action.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

Answer:

1. Production Data Representation

The production data can be represented using First-Order Predicate Calculus (FOPC) as:

Facts:

- $\text{Active}(\text{M1})$
- $\text{Active}(\text{M2})$
- $\text{Maintenance}(\text{M3})$
- $\text{Active}(\text{M4})$

Rules:

- $\forall x (\text{Active}(x) \rightarrow \text{Producing}(x))$
- $\forall x \forall y (\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y))$
- $\forall x (\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x))$

From the first rule:

- Producing(M1)
- Producing(M2)
- Producing(M4)

From the maintenance rule:

- $\neg$ Producing(M3)

2. Products Currently Available

The rule

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

means that a product **y** is available if an active machine **x** produces it.

However, the given production data does not specify which machines produce which products. Therefore, the available products cannot be determined completely from the information provided.

For example, if:

$\text{Produces}(\text{M1}, \text{Gear})$

and

$\text{Active}(\text{M1})$

then:

$\text{Available}(\text{Gear})$

3. Is Gear Production Affected by Maintenance?

M3 is under maintenance:

$\text{Maintenance}(\text{M3})$

Therefore:

$\text{Maintenance}(\text{M3}) \rightarrow \neg \text{Producing}(\text{M3})$

So, M3 is not producing while under maintenance.

If M3 produces Gear:

$\text{Produces}(\text{M3}, \text{Gear})$

then Gear production is affected because M3 cannot currently produce it.

Thus:

If M3 is the machine responsible for Gear production, Gear production is affected by maintenance.

4. Effectiveness of Predicate Logic

First-Order Predicate Calculus is effective for industrial decision-support because it:

- Represents machine states and production relationships clearly.
- Allows automatic inference from predefined rules.
- Helps identify unavailable machines and production impacts.
- Supports logical decision-making and fault analysis.
- Can be extended with additional machines, products, and rules.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

Answer:

1. Correct Word Sense for Each Query

User Query	Correct Sense	Reason
Apple accessories	Technology Brand	The clicked result is an iPhone Charger , which is related to Apple products.

Mouse wireless	Computer Device	The clicked result is a Bluetooth Mouse , indicating a computer accessory.
Java tutorial	Programming Language	The clicked result is Coding Lessons , which relates to Java programming.
Python course	Programming Language	The clicked result is Software Development Training , indicating Python programming.

2. Semantic Cues Used for Disambiguation

The main semantic cues are:

- **Clicked result:** Provides strong evidence about the user's intended meaning.
- **Related product terms:** "Accessories" and "iPhone Charger" indicate the technology meaning of Apple.
- **Domain-specific terms:** "Wireless" and "Bluetooth Mouse" indicate a computer device.
- **Context words:** "Tutorial" and "Course" strongly suggest programming languages rather than an island or animal.
- **User behavior:** Click history can reveal what meaning the user intended.

For example:

"Python course" + "Software Development Training" → Programming Language

3. Impact of Incorrect Sense Selection

Incorrect WSD can significantly reduce search-engine performance.

For example, if **"Apple accessories"** is interpreted as fruit, the search engine may recommend:

- Fruit baskets
- Apple juice
- Fruit-related products

instead of:

- iPhone chargers
- AirPods
- Apple cases

This can result in:

- Lower search relevance
- Lower click-through rate (CTR)
- Poor recommendations
- Increased user frustration
- Reduced conversion rate
- Loss of customer trust

Thus, accurate word-sense selection is important for both **search quality and e-commerce revenue**.

4. Industrial-Scale WSD Strategy

An e-commerce company can use a **hybrid contextual WSD system**:

User Query → Context Analysis → Candidate Senses → ML/NLP Model → Correct Sense → Search & Recommendation

The strategy can include:

1. **Contextual embeddings** – Use models such as BERT/Transformer models to understand the meaning of words based on surrounding context.
2. **Product catalog knowledge** – Connect ambiguous words with product categories and attributes.
3. **Search and click history** – Use previous queries, clicks, purchases, and browsing behavior.
4. **User personalization** – Consider the user's interests and previous interactions.
5. **Entity linking** – Map words such as *Apple*, *Java*, and *Python* to appropriate entities.
6. **Continuous learning** – Retrain the model using new search queries and user feedback.
7. **Confidence-based disambiguation** – When confidence is low, show results covering multiple possible meanings or ask for clarification.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.

4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Answer:

1. How Syntax Contributes to Semantic Role Assignment

Syntactic structure identifies the relationship between the words in a sentence. It helps determine **who performs an action, what is affected, and who receives it**.

Examples:

- **Doctor prescribed medicine to patient.**
 - Doctor → **Agent**
 - Medicine → **Object/Theme**
 - Patient → **Recipient**
- **Patient reported severe headache.**
 - Patient → **Agent/Experiencer**
 - Headache → **Symptom**
- **Nurse monitored patient continuously.**
 - Nurse → **Agent**
 - Patient → **Object/Theme**
- **Medicine reduced blood pressure.**
 - Medicine → **Cause/Agent**
 - Blood pressure → **Affected entity/Theme**

Thus, the syntactic roles such as **Subject, Verb, and Object** provide important information for assigning semantic roles.

2. Are the Assigned Semantic Roles Appropriate?

Some roles are correct, but the table contains an important error.

Entity	Given Role	Appropriate Role	Reason
Doctor	Agent	Agent	Doctor performs the action of prescribing.
Medicine	Instrument	Theme/Object	Medicine is what is prescribed; it is not an instrument in this sentence.
Patient	Recipient	Recipient	Patient receives the medicine.
Headache	Symptom	Symptom	Headache represents the patient's symptom.

For the remaining sentences:

- **Nurse → Agent**
- **Patient → Object/Theme**
- **Medicine → Cause/Agent**
- **Blood pressure → Theme/Patient**

Therefore, **Medicine** → **Instrument is incorrect** in "Doctor prescribed medicine to patient." It should be **Theme/Object**.

3. Errors Caused by Incorrect Parsing

Incorrect syntactic parsing can lead to incorrect semantic interpretation.

For example:

"Doctor prescribed medicine to patient."

If the system incorrectly identifies **patient** as the subject, it may interpret the patient as prescribing the medicine instead of receiving it.

Possible consequences in healthcare include:

- Incorrect patient medication records
- Wrong identification of treatment providers
- Incorrect extraction of symptoms
- Misinterpretation of medical events
- Errors in clinical decision-support systems
- Incorrect information in electronic health records

Healthcare applications require high accuracy because semantic errors can potentially affect clinical decisions.

4. Methods to Improve Syntax-Driven Semantic Analysis

1. **Use medical-specific NLP models** trained on clinical text.
2. **Use dependency parsing** to identify relationships between medical entities.
3. **Apply semantic role labeling (SRL)** to identify Agent, Patient, Theme, Symptom, etc.
4. **Use medical ontologies** such as SNOMED CT and UMLS for standardized medical concepts.
5. **Handle negation carefully**, e.g., "Patient has no headache."
6. **Use contextual models** such as Transformer/BERT-based clinical NLP models.
7. **Human validation** should be used for critical or uncertain cases.
8. **Confidence scoring** can flag ambiguous interpretations for review.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____